AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

Test Report No. 2023 – 0343

Agricultural Machinery – Walking-Type Agricultural Tractor WESTWIND STA 201

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Test Report Number	2023 – 0343
Date of Issuance	March 25, 2024
Valid Until	March 25, 2029
Agricultural Machinery	Walking-Type Agricultural Tractor
Type	Pull type, Single axle
Brand	WESTWIND
Model	STA 201
Test Requested by	ILOILO NEW AGRI-INDUSTRIAL, MARKETING AND GENERAL SERVICES INC. Lawaan Village, Balantang, Jaro, Iloilo

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 345:2022	AMTEC Test
Power Requirement of the Prime Mover, kW, maximum	13.5	5.22[b]
Actual Field Capacity, ha/h	ND[a]	0.150
Peak Transmission Efficiency, %, minimum	85	90.95
Breakdowns/malfunctions during the five-hour continuous running test.	Shall have none	None

[a] Not specified by the requesting party.
[b] Based on engine nameplate.
Note: ND – No Data

OBSERVATIONS

1. Operator's manual	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	2013 – 0202 and 2017 – 0171
3. Manufacturer	ATON MARKETING Gov. Villaverte St., San Jose, Antique
4. Prime mover used	5.22 kW KUBOTA RK70 Diesel Engine
5. Serial number	None
6. Number of operators	One
7. Handling and stability of the WTAT	The machine was stable during the test operation. The operator was able to maneuver the WTAT with ease during working and turning corners.
8. Manipulating of the operating levers	The main clutch lever is accessible to the left and right hand of the operator.
9. Replacing and adjusting of parts	Locally fabricated parts can be replaced. Belt tension can be adjusted through the movable engine base.
10. Safety features	Engine load can be immediately disengaged from the input shaft by releasing the main clutch lever.
11. Failure or abnormalities that may be observed on the WTAT or its component parts.	None
12. The WTAT shall be generally made of steel bars and sheet metals.	Steel materials were generally used in the fabrication of the machine.
13. The chain and sprocket used shall conform to PAES 303:2000 (Engineering materials - Roller chains and sprockets for agricultural machines – Specifications and applications) and shall have provisions for refilling of lubrication oil.	The material used in the chain and sprocket was not verified if it conformed to PAES 303:2000 due to limited technical capabilities.

OBSERVATIONS (Cont'n)

14. The hand grip shall be made of non-slip material and shall have a minimum nominal diameter of 25 mm.	The handle bar is made of steel pipe with no rubber grip. The outside diameter of the hand grip is 26.55 mm.
15. Mechanism for handle bar height adjustment with at least three settings shall be provided. The handle bar height shall be 650 mm to 800 mm.	The handle bar height is fixed at 740 mm from the ground.
16. The hitch of the WTAT shall be in accordance with the specifications of PAES 107:2000 (Agricultural machinery - Hitch for walking-type agricultural tractor – Specifications).	The WTAT has a one-hole hitch. The material used for the hitch was not verified due to limited technical capability. The specifications of the different components of the hitch are summarized in Page 9.
17. The speed control lever shall be provided and accessible to and operated from the perspective of the operator. It shall be provided with a detent mechanism capable of setting and retaining/locking the speed at any point within the range of the engine speed.	Not provided
18. The main clutch lever shall be provided and accessible to and operated from the perspective of the operator.	The main clutch levers are accessible to the left and right hand of the operator and can be operated independently.
19. Stand assembly for use when the WTAT is not in operation shall be provided.	Provided
20. There shall be earmuffs or other ear protection device provided for the operator to use when 95 dB(A) is exceeded during operation. Recommendations for use of ear protection device shall likewise be indicated in the operator's manual.	The maximum noise level measured during the field performance test was 92.1 dB(A). Earmuffs or other ear protection devices were not provided for the operator since the noise level did not exceed 95 dB(A).
21. The WTAT shall be free from any manufacturing defects that may be detrimental to its operation.	No manufacturing defects were observed during the test operation.
22. All metal surfaces shall be painted properly.	All parts were properly painted and free from rust.
23. Parts of the WTAT that are exposed to the operator shall be free from sharp edges and rough surfaces.	No sharp edges and surfaces were observed during the test operation.
24. Appropriate labels, safety symbols, and warning notices shall be provided in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	No safety symbols or warning notices on the WTAT.

OBSERVATIONS (Cont'n)

25. All controls shall be in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	The main clutch lever is accessible to the left and right hand of the operator. Markings and labels of the controls are not provided.
26. All exposed hot surfaces shall be provided with heat shield or protection to minimize the possibility of inadvertent contact with the operator.	The exhaust outlet of the prime mover was directed away from the operator. However, a heat shield or protection was not provided.
27. Mechanism for emergency stop of the engine and immediate load disengagement of the transmission system shall be provided.	The power transmission can be disengaged using the main clutch lever.
28. Mechanism for belt transmission adjustment shall be provided.	Belt transmission can be adjusted using the main clutch lever or by adjusting the engine base position.
29. Belt cover and mud guard shall be provided. All other moving parts shall have safety guards.	No belt cover was provided.
30. There shall be a provision for means to minimize vibration.	A mechanism to minimize vibration is not provided.
31. The hexagonal axle shall be in accordance with the specifications of PAES 108:2000 (Agricultural machinery – Hexagonal axle and hub for walking-type agricultural tractor – Specifications).	The hexagonal axle used is Type 2. The dimensions of the axle and axle hub are summarized in Pages 8-9. The fastener used is locking bolts.
32. The axle shall be able to accommodate compatible pneumatic tires provided for transport.	Not provided
33. Other remarks	1. The machine had no nameplate at the time of the test. 2. Disc plow and Spike-tooth harrow were available attachments of the machine. However, only the disc plow was used during the test operation.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Dates of Test	July 6, 2023, and March 21, 22, and 23, 2024
Location of Test	Brgy. Agcuyawan Calsada, Barotac Nuevo, Iloilo 10°53'26.53" N, 122°43'36.43" E; and AMTEC Bldg., WESVIARC, Brgy. Buntatala, Jaro, Iloilo City, Iloilo
Items Tested	Operator's Manual, Width of Tillage, Depth of Tillage, Field Capacity, Field Efficiency, Noise Level, Number of Operators, Varying Load Performance Test, Continuous Running Test, Transmission Efficiency Test, and Fuel Consumption Rate

METHODS OF TEST

The WTAT was tested based on the PNS/BAFS 348:2022 Agricultural Machinery – Walking-Type Agricultural Tractor – Methods of Test.

DESCRIPTION OF THE MACHINE

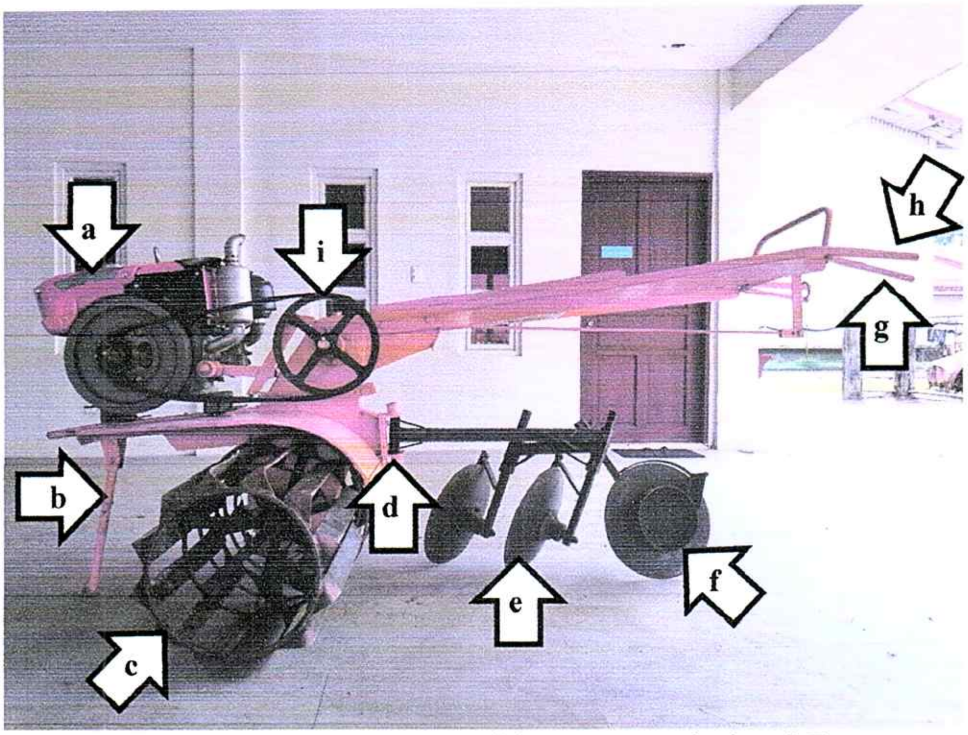

Figure 1. Parts of WESTWIND STA 201 Walking-Type Agricultural Tractor powered by 5.22 kW KUBOTA RK70 Diesel Engine include the following: (a) Prime mover, (b) Stand, (c) Cage wheel, (d) One-hole Hitch point, (e) Disc plow, (f) Rear furrow wheel, (g) Main clutch lever, (h) Handle, and (i) V-belt and pulley transmission system.

Figure 2. Laboratory test set-up of the WESTWIND STA 201 Walking-Type Agricultural Tractor.

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	2660	2660
1.1.2	Width	1260	1275
1.1.3	Height	970	1165
1.1.4	Ground clearance	110	145
1.2	Weight, kg	ND	132
2	Engine		
2.1	Brand	KUBOTA	KUBOTA[a]
2.2	Model	RK70-S	RK70-S[a]
2.3	Make or manufacturer	Indonesia	Indonesia[a]
2.4	Serial number	RK70-APG0129	RK70-APG0129[a]
2.5	Type	ND	Four-stroke, single horizontal cylinder, diesel engine
2.6	Rated power, kW	5.22	5.22[a]
2.7	Rated speed, rpm	2400	2400[a]
2.8	Displacement, cm^3	376	376[a]
2.9	Cooling system	Water-cooled	Water-cooled
2.10	Starting system	Manual cranking	Manual cranking lever
3	Type of clutch		
3.1	Main clutch	ND	Idler clutch
3.2	Steering clutch	ND	None
3.3	Tilling clutch	ND	None
4	Power transmission system		
4.1	Engine to input	Double groove B-section	Belt and pulley
4.2	Engine[b]	102 × 2B × 30	101 × 2B × 29.29
4.3	Input[b]	305 × 2B × 30	301 × 2B × 28.26
4.4	Belt size	ND	B-73
5	Axle		
5.1	Type	Hexagonal, solid	Hexagonal
5.2	Overall length of shaft, mm	510	510
5.3	Width across the flat, mm	32	32.14
5.4	Diameter, mm	36	NA

[a] The data were obtained from the engine nameplate and the operator's manual provided by the requesting party.
[b] Pulley diameter, mm × number of belts × shaft diameter, mm
Note: NA – Not Applicable, ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
6	Hexagonal hub		
6.1	Length, mm	ND	205
6.2	Width across the flat, mm	ND	32.84
6.3	Thickness, mm	ND	9.88
7	Hitch point		
7.1	Type	One-hole	One-hole
7.2	Dimensions, mm[b]		
7.2.1	H	ND	120.18
7.2.2	h	ND	33.61
7.2.3	b	ND	36.64
7.2.4	c	ND	22.23
7.2.5	l	ND	95.77
7.2.6	d_i	ND	18.91
7.2.7	d_o	ND	38.18
7.2.8	t	ND	7.45
7.2.9	r	ND	NA

[a] The requesting party did not provide complete machine specifications.
[b] Dimensions based on PAES 107:2000 Table 1 – Dimensions of One-hole Hitch for Walking-type Agricultural Tractor
Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
8	Hitch pin		
8.1	d_p	ND	17.74
8.2	l_p	ND	208.00
8.3	l_s	ND	NA
9	Tractive wheels		
9.1	Pneumatic wheel	ND	None
9.2	Cage wheel		
9.2.1	Dimension, D × W, mm	525 × 555	500 × 540
9.2.2	Number of blades	ND	12
9.2.3	Number of flanges	NA	NA
9.2.4	Blade thickness	ND	5.74
9.2.5	Blade shape	ND	Rectangular
10	Attachments		
10.1	Disc plow		
10.1.1	Type	2-bottom disc plow	Plain
10.1.2	Number of discs	2	2
10.1.3	Diameter, mm	370	370
10.1.4	Thickness, mm	ND	4.67
10.1.5	Concavity, mm	ND	39.63
10.1.6	Disc spacing, mm	ND	265
10.1.7	Operating width, mm	ND	470
10.1.8	Material	ND	Steel
10.2	Harrow		
10.2.1	Type	Comb-tooth	Spike tooth
10.2.2	Number of tines	11	12
10.2.3	Tine diameter, mm	ND	15.14
10.2.4	Tine length, mm	ND	240
10.2.5	Material	ND	Plain round bars
10.2.6	Operating width, mm	1285	1190
10.3	Rotary tiller	NA	NA
11	Safety devices	ND	None
12	Special features	ND	None

[a] The requesting party did not provide complete machine specifications.
Note: NA – Not Applicable, ND – No Data

RESULTS

Item		Data
1	Test conditions	
1.1	Condition of field	
1.1.1	Type	Lowland
1.1.2	Location	Brgy. Agcuyawan, Barotac Nuevo, Iloilo
1.1.3	Soil type	Sta. Rita Clay Loam
1.1.4	Dimensions of field	
1.1.4.1	Length, m	32
1.1.4.2	Width, m	16
1.1.4.3	Area, m^2	512
1.1.5	Depth of water, mm	47
1.1.6	Number of days soaked	1
1.1.7	Soil resistance, kg/cm^2	NM[a]
1.1.8	Spacing of stubbles, rows × hills, mm	Random
1.1.9	Height of stubbles, mm	375
1.1.10	Weed density	Medium
1.2	Condition of the WTAT	
1.2.1	Implement used	
1.2.1.1	Type	Disc plow
1.2.1.2	Operating width, mm	470

[a] No available test instrument at the time of test.
Note: NM – Not Measured

RESULTS (Cont'n)

Item		Data
2	Field performance	
2.1	Type of field operation	Primary tillage
2.2	Field operational pattern	Circuitous rounded corners
2.3	Traveling speed, kph	3.27
2.4	Average depth of tillage, mm	90
2.5	Average width of tillage, mm	507
2.6	Theoretical width of tillage, mm	470
2.6	Actual field capacity	
2.6.1	ha/h	0.150
2.6.2	h/ha	6.73
2.7	Theoretical field capacity	
2.7.1	ha/h	0.154
2.7.2	h/ha	6.73
2.8	Field efficiency, %	97.27
2.9	Fuel consumption	
2.9.1	L/h	1.62
2.9.2	L/ha	10.84
2.10	Noise at operator's ear level, dB(A)	92.1
2.11	Unplowed, %	6.00

RESULTS (Cont'n)

Varying Load Performance Test

Table 1. Condition during the varying load performance test of WESTWIND STA 201 Walking-Type Agricultural Tractor.

Date of Test	March 21, 2024		
Place of Test	AMTEC Bldg., WESVIARC, Brgy. Buntatala, Jaro, Iloilo City, Iloilo		
Prime Mover Used	5.22 kW KUBOTA RK70-S Diesel Engine		
Test Conditions			
Dry Bulb Temperature, °C	34.4		
Relative Humidity, %	43.0		
No Load Speed, rpm	Prime Mover	Input	Axle
	2407	790	84
Speed Reduction Ratio	9.40:1		

Table 2. Results of the varying load performance test of WESTWIND STA 201 Walking-Type Agricultural Tractor.

Axle Load, kg	Speed, rpm			Belt Slippage, %	Axle Torque, N-m	Axle Shaft Power, kW	Fuel Consumption, L/h	Specific Fuel Consumption, g/kW-h	Temperature, °C	
	Engine Shaft	Input Shaft	Axle Shaft						Engine Exhaust	Engine Oil
0.612	2407	790	84	0	2.40	0	0.542	21589.97	78.4	42.9
13.10	2390	784	83	0.124	51.42	0.444	0.620	1173.44	80.3	54.6
25.37	2367	777	82	0.004	99.56	0.853	0.695	684.37	100.9	62.5
38.36	2351	772	82	0.072	150.53	1.292	0.755	490.75	104.3	71.6
51.02	2345	770	81	0.040	200.22	1.701	0.846	417.71	110.3	72.4
63.78	2340	768	81	0.094	250.27	2.13	0.942	372.12	126.3	77.1
77.04	2334	766	81	0.055	302.32	2.56	1.050	344.26	139.8	79.2
89.29	2326	757	80	0.874	350.36	2.95	1.089	310.48	175.9	80.6
102.19	2322	755	80	1.003	400.98	3.36	1.101	275.37	190.3	81.2
115.46	2316	753	80	0.980	453.07	3.79	1.47	324.72	250.9	82.5
127.29	2313	749	79	1.408	499.47	4.15	1.71	346.17	260.3	83.6

RESULTS (Cont'n)

Continuous Running Test

Table 3. Condition during the continuous running test of WESTWIND STA 201 Walking-Type Agricultural Tractor.

Date of Test	November 17, 2023		
Place of Test	AMTEC Bldg., WESVIARC, Brgy. Buntatala, Jaro, Iloilo City, Iloilo		
Prime Mover Used	5.22 kW KUBOTA RK70-S Diesel Engine		
Test Conditions			
Dry Bulb Temperature, °C	34.5		
Relative Humidity, %	57.3		
Duration of Test, h	5.0		
No Load Speed, rpm	Prime Mover	Input	Axle
	2200	720	76
Speed Reduction Ratio	9.40:1		

Table 4. Results of the continuous running test of WESTWIND STA 201 Walking-Type Agricultural Tractor.

Time, h	Axle Load, kg	Speed, rpm			Belt Slippage, %	Axle Torque, N-m	Axle Shaft Power, kW	Fuel Consumption, L/h	Specific Fuel Consumption, g/kW-h	Temperature, °C	
		Engine Shaft	Input Shaft	Axle Shaft						Engine Exhaust	Engine Oil
0	108.49	2162	703	75	1.00	425.72	3.33	1.22	308.34	202.5	63.5
0.5	108.96	2117	690	74	0.72	427.57	3.31	1.18	300.20	210.6	100.1
1.0	109.92	2123	688	74	1.25	431.31	3.34	1.19	298.78	217.6	101.0
1.5	109.80	2122	689	74	1.16	430.84	3.32	1.19	301.75	214.1	100.4
2.0	109.30	2122	689	74	1.18	428.90	3.31	1.18	300.62	229.7	101.1
2.5	109.67	2120	686	73	1.50	430.36	3.29	1.19	304.44	230.1	101.7
3.0	109.65	2127	689	74	1.30	430.27	3.33	1.19	300.69	227.6	101.4
3.5	109.63	2123	689	74	1.14	430.20	3.33	1.20	303.15	227.8	101.2
4.0	109.81	2128	691	74	1.15	430.88	3.34	1.17	295.57	226.4	101.2
4.5	109.82	2127	689	74	1.28	430.93	3.34	1.19	298.74	215.3	101.3
5.0	109.97	2116	685	73	1.36	431.50	3.31	1.16	294.46	277.6	102.1
Ave.	**109.55**	**2126**	**690**	**74**	**1.19**	**429.86**	**3.32**	**1.19**	**300.61**	**225.4**	**97.7**

RESULTS (Cont'n)

Transmission Efficiency Test

Table 5. Condition during the transmission efficiency test of WESTWIND STA 201 Walking-Type Agricultural Tractor.

Date of Test	March 23, 2024	
Place of Test	AMTEC Building, WESIARC, Brgy. Buntatala, Jaro, Iloilo City, Iloilo	
Prime Mover Used	5.97 kW KUBOTA RT80-1S Diesel	
Test Conditions		
Dry Bulb Temperature, °C	34.2	
Relative Humidity, %	65.7	
No Load Speed, rpm	Input shaft	Axle shaft
	789	84
Speed Reduction Ratio	9.40:1	

Table 6. Results of the transmission efficiency test of WESTWIND STA 201 Walking-Type Agricultural Tractor.

Axle Load, kg	Axle Torque, N-m	Speed, rpm		Axle Shaft Power, kW	Input Transmission Shaft		Efficiency, %
		Input Shaft	Axle Shaft		Torque, kg-m	Power, kW	
0.50	1.95	789	84	0.017	0.17	0.140	12.22
12.92	50.70	783	83	0.438	0.75	0.602	72.74
25.85	101.43	775	82	0.870	1.29	1.03	84.57
38.63	151.58	773	82	1.30	1.92	1.52	85.39
51.37	201.56	770	82	1.72	2.46	1.95	88.48
64.16	251.76	768	82	2.15	3.07	2.42	88.69
76.85	301.55	761	81	2.56	3.68	2.87	88.95
89.47	351.07	755	81	2.96	4.27	3.31	89.39
102.35	401.62	751	80	3.37	4.83	3.73	90.45
114.67	449.97	747	80	3.77	5.40	4.14	90.95
127.54	500.47	744	79	4.14	6.05	4.62	89.65

OBSERVATIONS

Figure 3. Nameplate of the prime mover, markings, and labels of the WESTWIND STA 201 Walking-Type Agricultural Tractor.

Tested by:

FRANKLIN C. PAJARON
Test Engineer

Verified by:

for
KAREN CYBILL F. CABANGLAN
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO
AMTEC Director

MAR 26 2024
Date

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